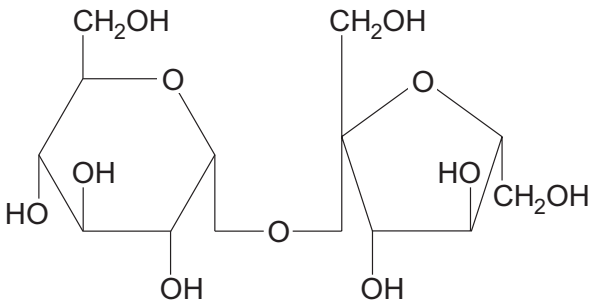


4. Sucrose is a disaccharide of glucose and fructose. The enzyme sucrase, catalyses the hydrolysis of sucrose into its monosaccharides. A colorimeter can be used to record absorbance values, which can then be used to determine the rate of hydrolysis.
- (a) (i) Complete the diagram to show the hydrolysis of sucrose including the products formed. [2]



(ii) State the name of the bond broken during the reaction. [1]

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(iii) Explain why glucose and fructose are referred to as structural isomers. [1]

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(iv) A student was provided with two beakers, one containing sucrose and the other containing sucrase. Describe **one** biochemical test that the student could have carried out to distinguish between the two solutions. [2]

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(b) A student wanted to investigate the hypothesis that sucrase catalyses the hydrolysis of sucrose fastest at a neutral pH. She was provided with the following method:

- Add 5 cm³ of buffer solution to a test tube
- Add 5 cm³ of sucrose solution to the test tube
- Add 1 cm³ of sucrase solution to the test tube and mix
- After 20 minutes add 1 cm³ of dinitrosalicylic acid (DNS)
- Pipette 5 cm³ of the mixture into a cuvette and place into a colorimeter and record the absorbance of light passing through the solution.

Dinitrosalicylic acid (DNS) will react with monosaccharides to produce amino-nitrosalicylic acid (ANS). ANS causes a colour change to occur which can be detected by a colorimeter. The greater the concentration of ANS the greater the absorbance of light.

(i) State **two** additional variables that the student should control to ensure that the results recorded would be repeatable. [2]

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(ii) The results are shown in the table below.

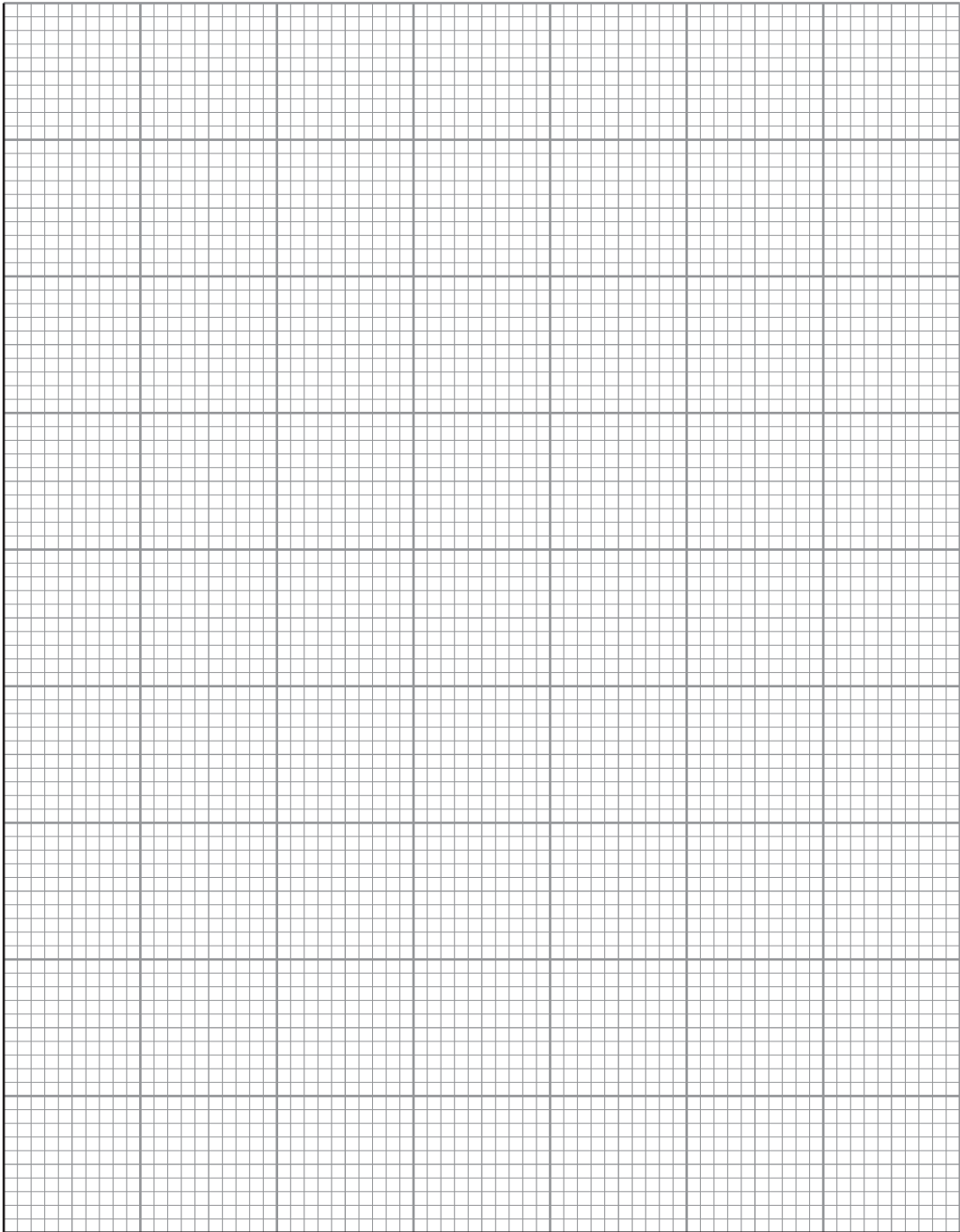
pH of sucrose and sucrase solution	Absorbance of light (au)			
	trial 1	trial 2	trial 3	mean
3	0.84	0.63	0.76	0.74
4	0.89	0.95	0.76	0.87
5	0.85	0.91	0.72	0.83
7	0.27	0.34	0.29	0.30
9	0.13	0.11	0.12	0.12
10	0.05	0.04	0.06	0.05



Plot the mean data on the graph below.

[4]

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(iii) What conclusions can be drawn from the data regarding the hypothesis being tested? [3]

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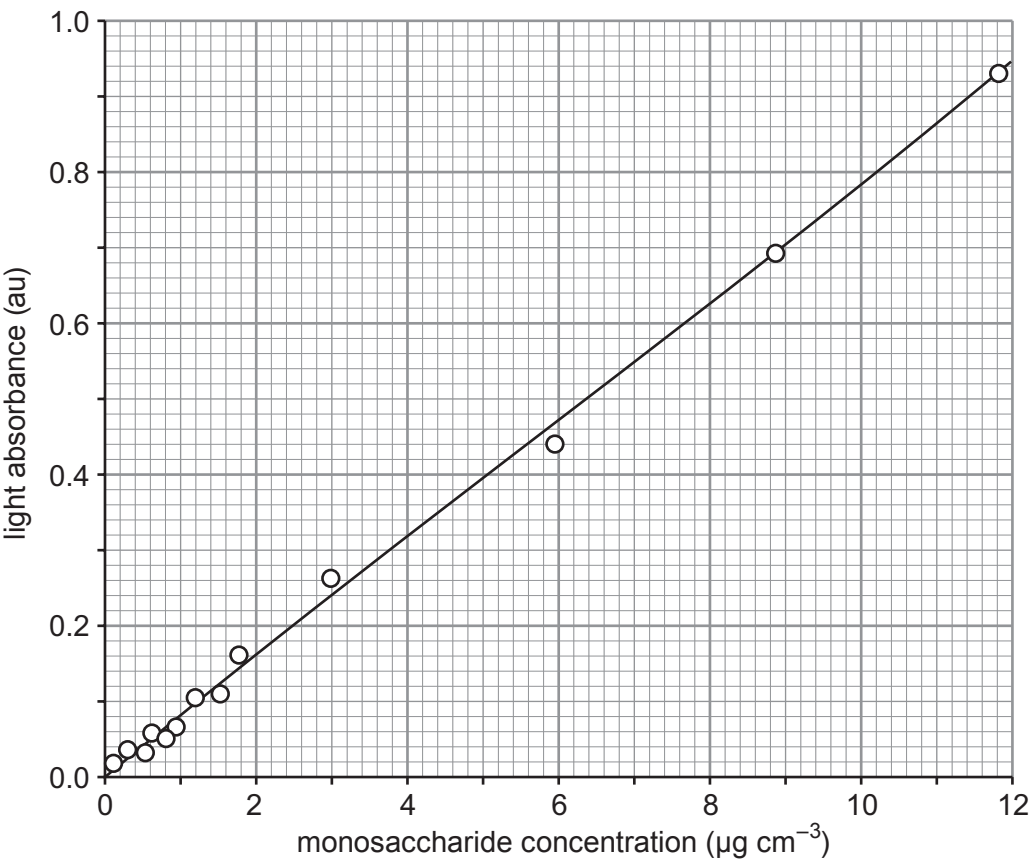
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(iv) The student was also provided with the calibration curve below in order to calculate the concentration of glucose produced during the investigation.

The colour change, caused by ANS, is dependent on the concentration of monosaccharides, the greater the concentration the greater the absorbance value recorded.



Determine the maximum concentration of glucose produced during the investigation at pH 7. [2]

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(c) With reference to the data explain how, and why, you would modify the method to determine the optimum pH more accurately. [2]

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